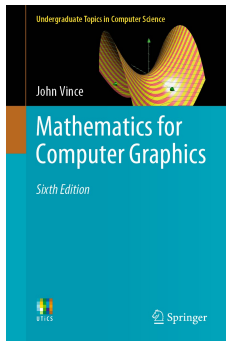


Refresh on Mathematics

Introduction to Computer Graphics - Lecture 02

Pirmin Pfeifer



Mathematics for Computer
Graphics
John Vince

[Download PDF](#)

Related Chapters:

- ▶ 3 Algebra
- ▶ 4 Trigonometry
- ▶ 5 Coordinate Systems
- ▶ 7 Vectors

Why do we need Mathematics for CG?



Computer Graphics are inherently mathematical.
To render images we need a good foundation of Mathematics.¹

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- ▶ Coloring and Shading → more Vector-Mathematics
- ▶ Animation → Trigonometry and more Matrices

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Which Mathematics do we need?

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Relevant topics include but are not limited to:

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- ▶ Number-Systems (hex,oct,binary)



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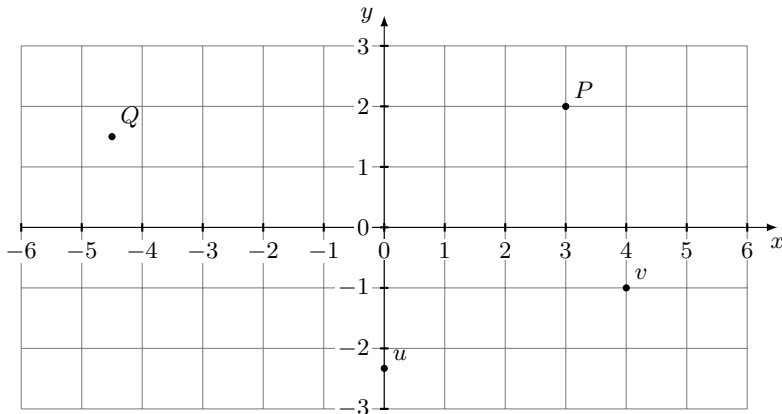
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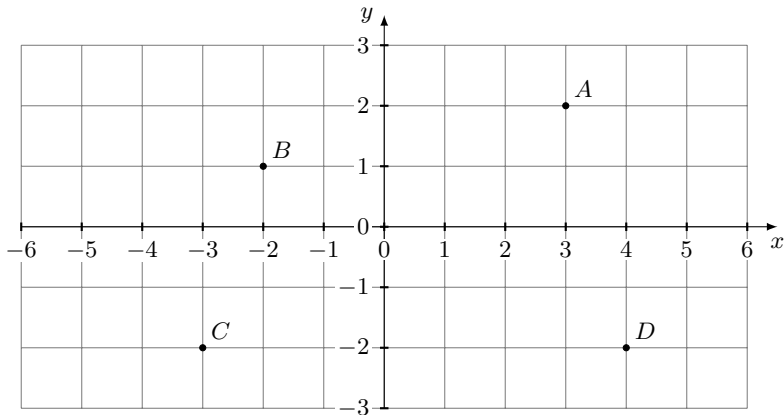
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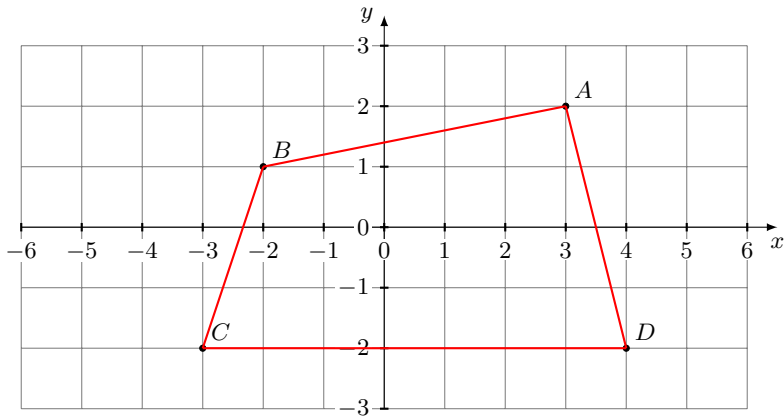
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- ▶ Vectors & Matrices

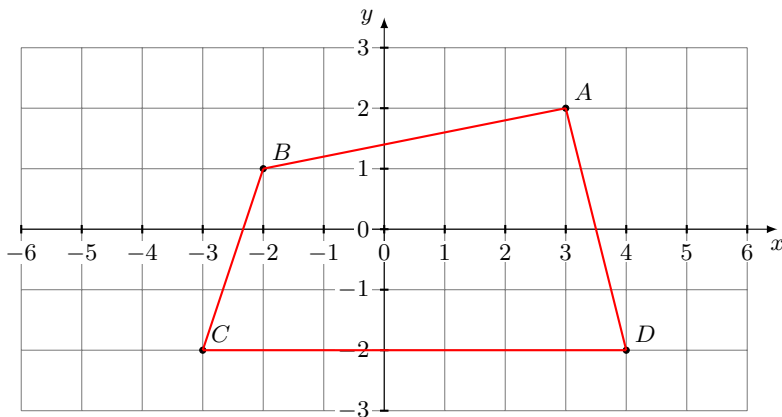


Points consist out of x and y value. Coordinates of P are $(3, 2)$.



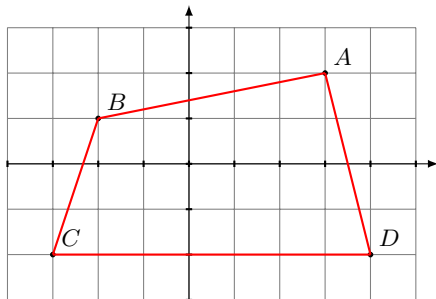


We define the Polygon $P = (A, B, C, D)^2$.

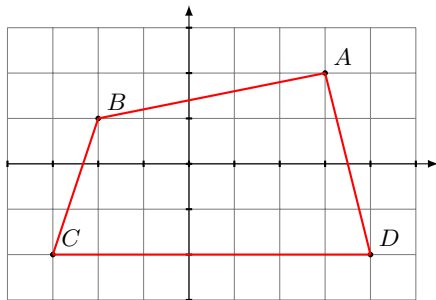


We define the Polygon $P = (A, B, C, D)$ ².

²The order is important (in this case the order is counterclockwise)



x	y
$x_A = 3$	$y_A = 2$
$x_B = -2$	$y_B = 1$
$x_C = -3$	$y_C = -2$
$x_D = 4$	$y_D = -2$

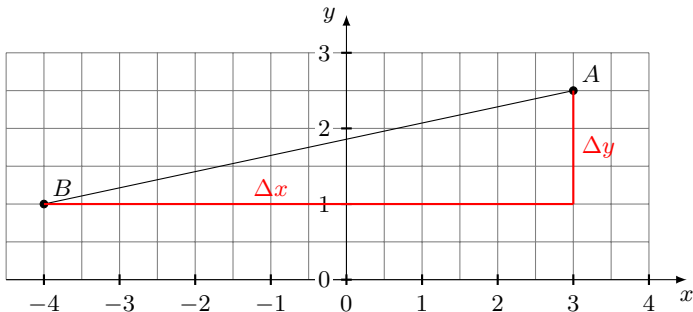


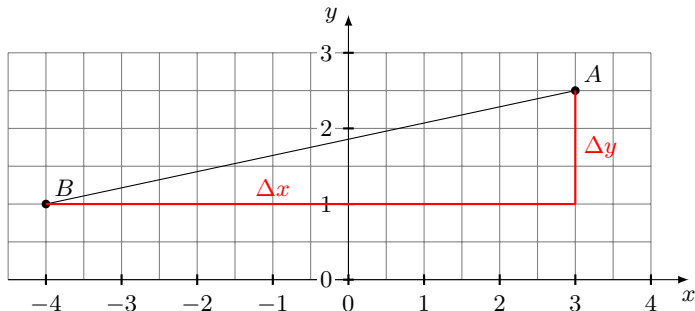
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Area of a 2D Polygon

area =

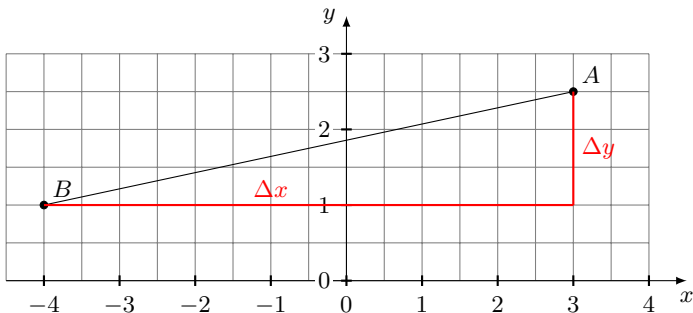
$$\frac{1}{2}((x_A y_B - x_B y_A) + (x_B y_C - x_C y_B) + (x_C y_D - x_D y_C) + (x_D y_A - x_A y_D))$$





Theorem of Pythagoras

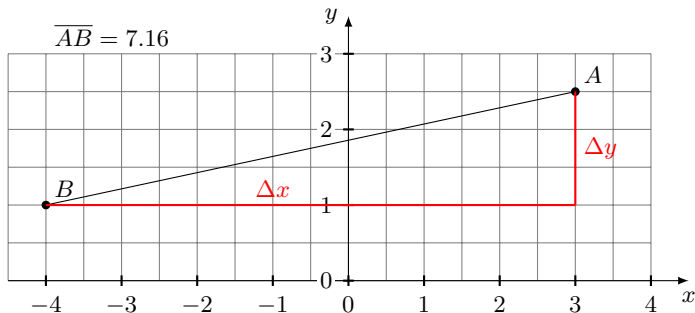
$$distance = |AB| = \sqrt{(\Delta x)^2 + (\Delta y)^2}$$



Theorem of Pythagoras

$$\text{distance} = |\overline{AB}| = \sqrt{(\Delta x)^2 + (\Delta y)^2}$$

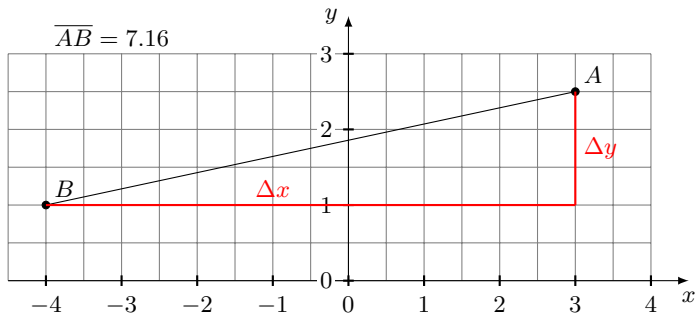
$$\text{Distance } \overline{AB} = \sqrt{(3 - (-4))^2 + (2.5 - 1)^2}$$



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$$\text{Distance } \overline{AB} = \sqrt{(3 - (-4))^2 + (2.5 - 1)^2} = \sqrt{7^2 + 1.5^2} \approx 7.16$$



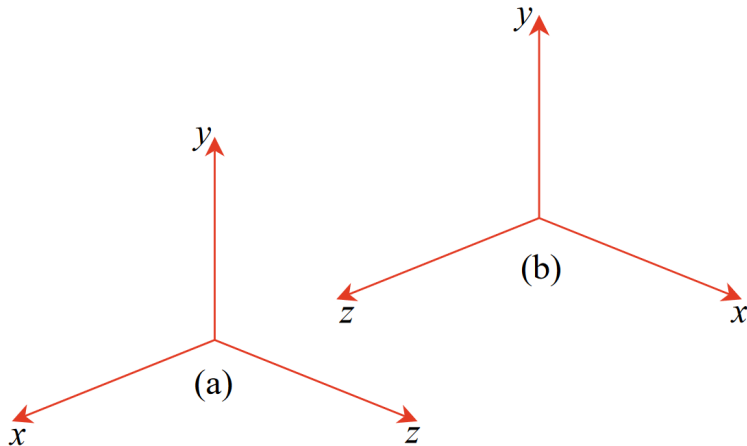
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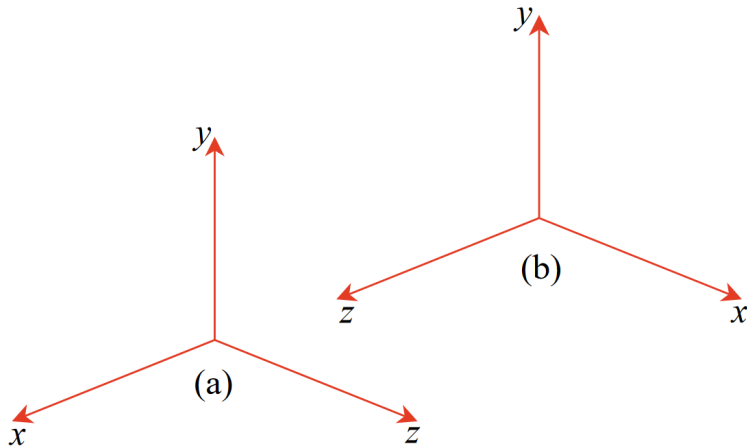
3D Coordinates Systems

Extends the 2D coordinate system with a z dimension



3D Coordinates Systems

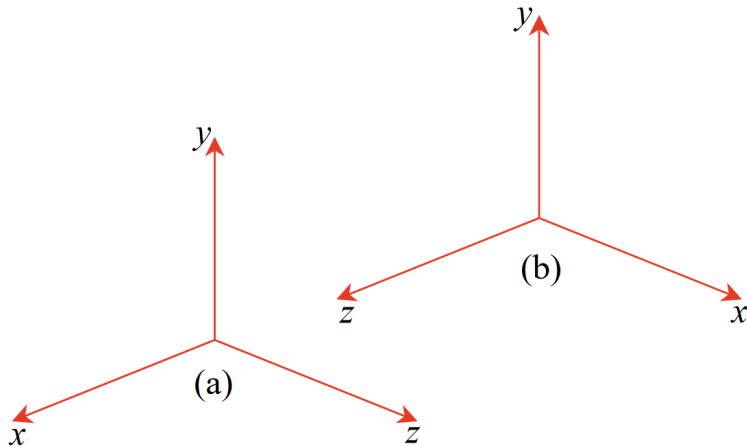
Extends the 2D coordinate system with a z dimension



(a) := left-handed axial system

3D Coordinates Systems

Extends the 2D coordinate system with a z dimension



(a) := left-handed axial system

(b) := right-handed axial system

Theorem of Pythagoras - 2D

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Theorem of Pythagoras - 3D

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Generic Vectors

Ordered collection off values of same type

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Three values commonly named (x, y, z) or (r, b, g)

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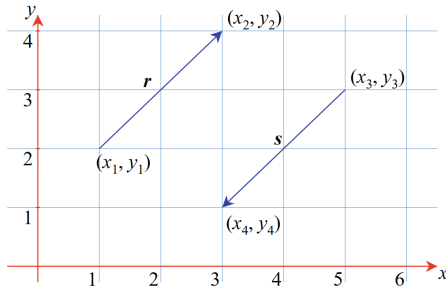
Vector 3D (*int3, uint3, float3, glm :: vec3*)

Three values commonly named (x, y, z) or (r, b, g)

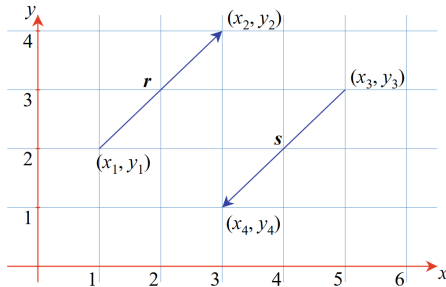
Vector 4D (*int4, uint4, float4, glm :: vec4*)

Four values commonly named (x, y, z, w) or (r, g, b, a)

Vectors are commonly associated with points in space or directions.



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Magnitude of a Vector

$$||r|| = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}$$

Scaling a vector

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$$n = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad \text{then} \quad \lambda n = \begin{bmatrix} \lambda x \\ \lambda y \\ \lambda z \end{bmatrix}, \quad \text{where} \quad \lambda \in \mathbb{R}.$$

Vector Addition and Subtraction

$$r = \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix}, \quad s = \begin{bmatrix} x_s \\ y_s \\ z_s \end{bmatrix}$$

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Addition is commutative

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Subtraction is **not** commutative

$$r - s = \begin{bmatrix} x_r - x_s \\ y_r - y_s \\ z_r - z_s \end{bmatrix} \neq s - r$$

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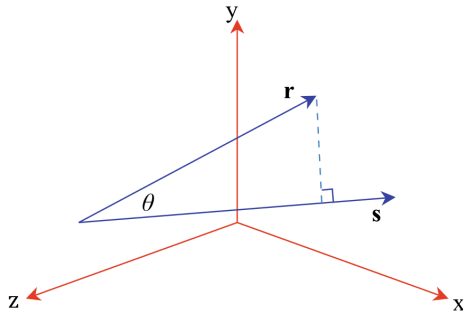
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³Unless the magnitude is equal 0

Scalar⁴/Dot Product

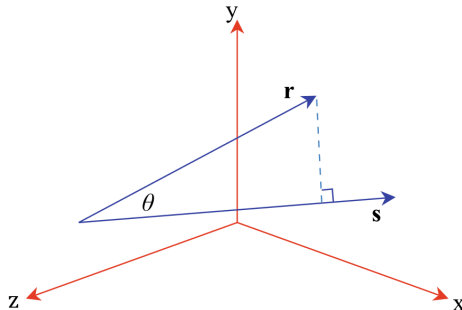


⁴It is called scalar product because it computes a scalar (one dimensional) value.

Scalar⁴/Dot Product

Scalar Product:

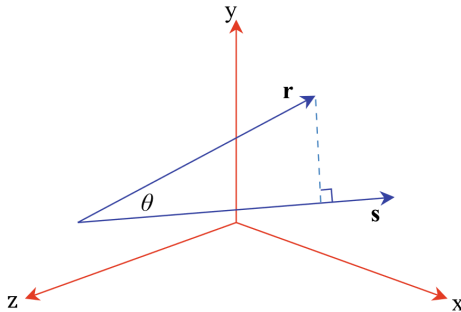
$$r \cdot s = ||r|| \cdot ||s|| \cdot \cos \theta$$



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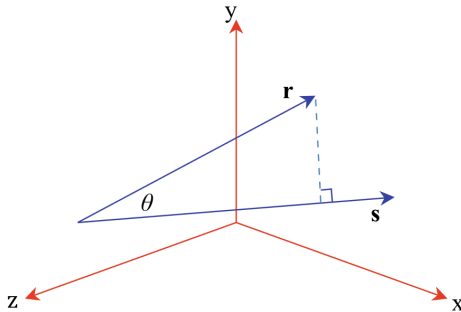


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Scalar Product:

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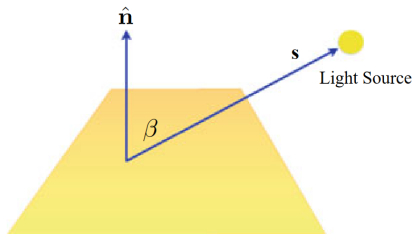


The Scalar Product can be extended to:

$$r \cdot s = ||r|| \cdot ||s|| \cdot \cos \theta = x_r x_s + y_r y_s + z_r z_s$$

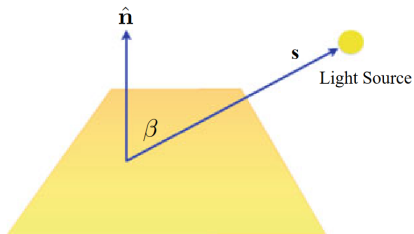
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Dot Product - Examples

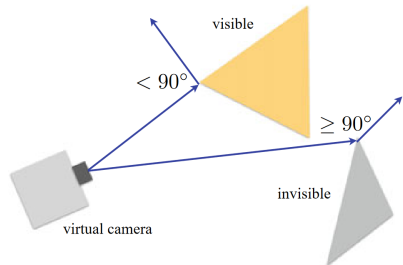


Lighting calculations

Dot Product - Examples



Lighting calculations



Backface detection

Cross Product

The cross product calculates a new vector t :

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which is the area of the parallelogram formed by r and s :

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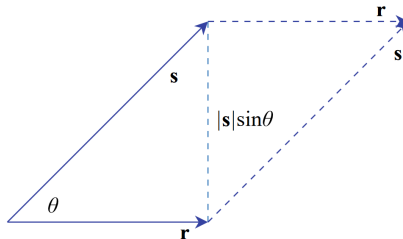
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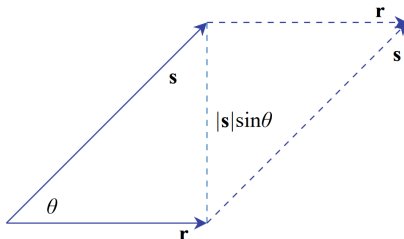


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The resulting vector t is perpendicular to the parallelogram⁵.

⁵In 3 and higher dimensions

Cross Product

The cross product calculates a new vector t :

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The cross product can be extend to:

$$\begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} \times \begin{bmatrix} x_s \\ y_s \\ z_s \end{bmatrix} = \begin{bmatrix} y_r z_s - z_r y_s \\ z_r x_s - x_r z_s \\ x_r y_s - y_r x_s \end{bmatrix} = \begin{bmatrix} x_t \\ y_t \\ z_t \end{bmatrix}$$

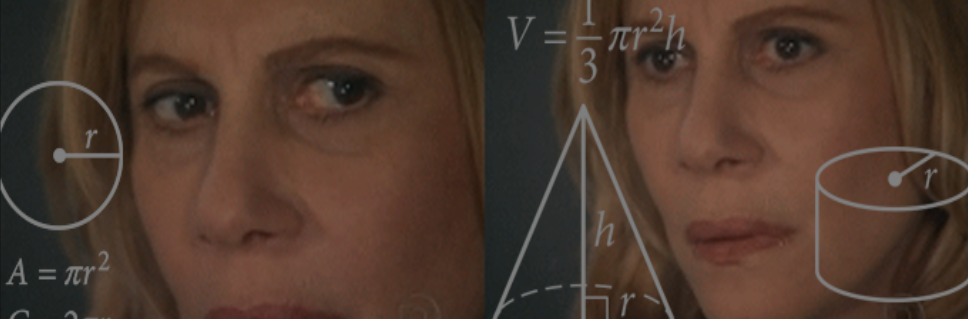
Conclusion

Today we (re)learned:

- ▶ logarithms
- ▶ trigonometry
- ▶ coordinate systems
- ▶ vectors

We still need a bit more math, but we will look at it in a later lecture:

- ▶ Matrices
- ▶ Affine Transforms
- ▶ Interpolation
- ▶ Curves
- ▶ Barycentric Coordinates



Any Questions?

